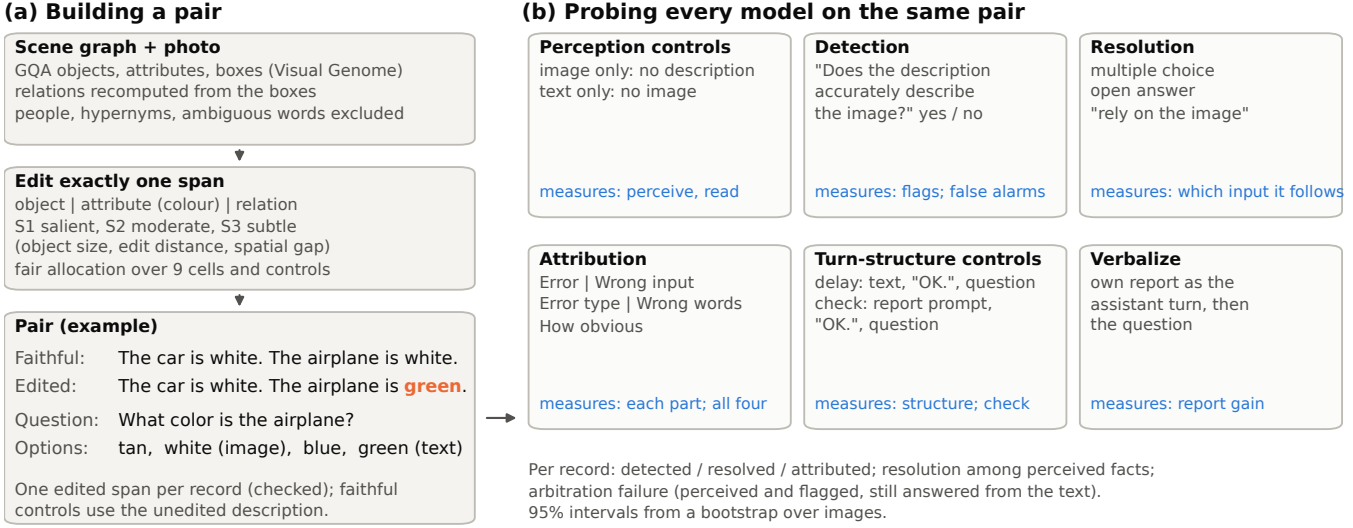
(a) Building a pair
Scene graph + photo
GQA objects, attributes, boxes (Visual Genome)
relations recomputed from the boxes
people, hypernyms, ambiguous words excluded
Edit exactly one span
object | attribute (colour) | relation
S1 salient, S2 moderate, S3 subtle
(object size, edit distance, spatial gap)
fair allocation over 9 cells and controls
Pair (example)
Faithful: The car is white. The airplane is white.
Edited: The car is white. The airplane is green.
Question: What color is the airplane?
Options: tan, white (image), blue, green (text)
One edited span per record (checked); faithful controls use the unedited description.
(b) Probing every model on the same pair
Perception controls
image only: no description
text only: no image
measures: perceive, read
Detection
"Does the description accurately describe the image?" yes / no
measures: flags; false alarms
Resolution
multiple choice
open answer
"rely on the image"
measures: which input it follows
Attribution
Error | Wrong input
Error type | Wrong words
How obvious
measures: each part; all four
Turn-structure controls
delay: text, "OK.", question
check: report prompt, "OK.", question
measures: structure; check
Verbalize
own report as the assistant turn, then the question
measures: report gain
Per record: detected / resolved / attributed; resolution among perceived facts;
arbitration failure (perceived and flagged, still answered from the text).
95% intervals from a bootstrap over images.